Getting to Know Plant Simulation

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Introduces you to the basic concepts on which simulation is based and to the basics of working with *Plant Simulation*.

Compare Modeling the Material Flow

Compare Visualizing the Material Flow

Simulation and Modeling Concepts

Simulation and Modeling Concepts

Simulation concepts and modeling concepts introduce you to the theoretical background of simulation as such and show you what to keep in mind before you start modeling.

In general, operations research processes are intended to allow you to make the right decisions, qualitatively as well as quantitatively. They formulate optimization models, containing all relevant factors, such as target function, conditions and target description. These processes require a large amount of processing power the more detailed the model is. Besides, the results and acceptance of operations research processes often are not satisfactory.

In addition to linear optimization models **simulation** nowadays is increasingly used for making the right decisions. Simulation offers good solutions for complex problems but does not automatically create the actual optimum. This is justified by the comparatively low amount of mathematical expenditure to obtain that result.

As processes to be analyzed become more complicated and complex and as more factors have to included, the more important simulation becomes with its analysis of real processes. These processes cannot be covered by mathematical solution processes or optimization processes or they may be realized only by using a large amount of resources.

The aim of simulation is to arrive at objective decisions by dynamic analysis, to allow managers to safely plan and, in the end, to reduce costs.

Thus, if real systems and plants are too expensive for conducting experiments and the time to conduct trials is too limited and too expensive, modeling, simulation and animation are excellent tools for analyzing and optimizing time dynamic processes.

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Go to Using Inheritance

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What is Simulation?

VDI Technical Rule 3633 defines simulation as the emulation of a system, including its dynamic processes, in a model one can experiment with. It aims at achieving results that can be transferred to a real world plant. In addition, simulation defines the preparation, execution and evaluation of carefully directed experiments within a simulation model.

As a rule, you will execute a simulation study like this:

- You first check out the real-world plant you want to model and collect the data you need for creating your simulation model.
- You then abstract this real-world plant and create your simulation model according to the aims of the simulation studies.
- After this, you run experiments, i.e., execute simulation runs, within the simulation model. This will produce a number of results, such as how often machines fail, how often they are blocked, which set-up times accrue for the individual types of station, which utilization the machines have, etc.
- The next step will be to interpret the data the simulation runs produce.
- Finally, management will use the results as a base for its decisions about optimizing the real plant.

Developing your simulation model is a cyclical and evolutionary process. You will start out with a first draft of your model and then refine and modify it to make use of the intermediary results the simulation runs provide. Eventually, after several cycles, you will arrive at your final model.

As a simulation expert, you must never loose sight of these questions:

- What do you want to accomplish with the simulation study?
- What are you examining?
- Which conclusions do you draw from the results of the simulation study?
- How do you transfer the results of the simulation study to the real-world plant?

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Time-Oriented Simulation and Event-Controlled Simulation

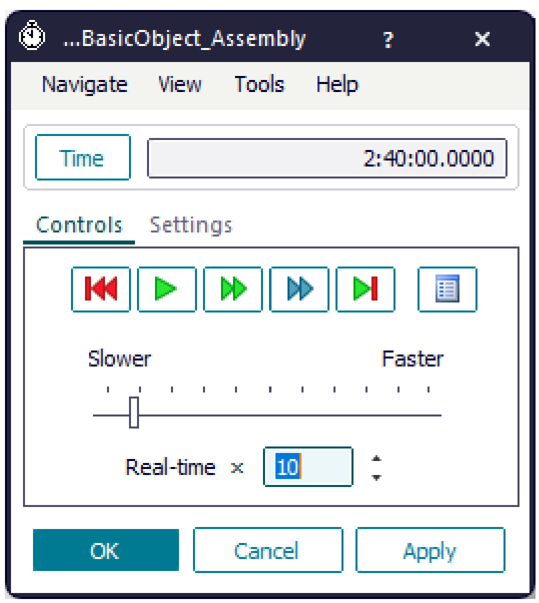
Plant Simulation is a discrete, event-controlled simulation program, i.e., it only inspects those points in time, at which events take place within the simulation model.

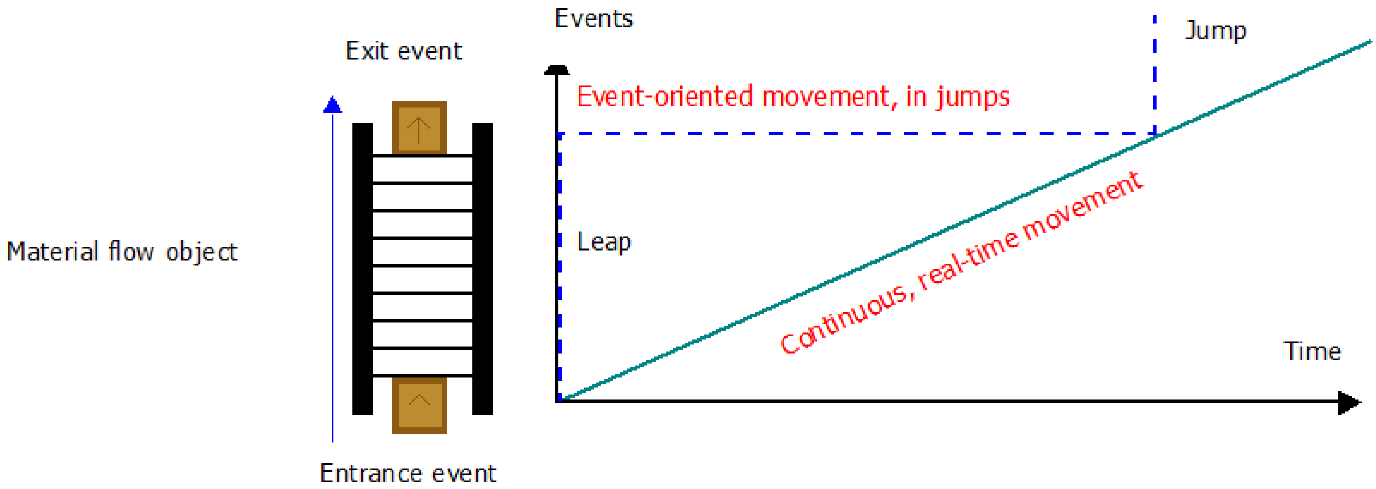
In reality, on the other hand, time elapses continually. When watching a part move along a conveyor system, you will detect no jumps in time. The curve for the distance covered, and the time it takes to cover it, is continuous, it is a straight line.

A discrete, event-controlled simulation program on the other hand only takes points in time (events) into consideration that are important to the further course of the simulation. Such events may, for example, be a part entering a station or leaving it or of it moving on to another machine. Any movements in between are of little interest to the simulation as such. It is only important that the entrance and the exit (**Out**) events are displayed correctly. When a part enters a material flow object, *Plant Simulation* computes the time until it exits that object and enters an exit event into the list of scheduled events of the *EventController* for this point in time.

Thus, the simulation time that the *EventController* displays, jumps from event to event. This happens as soon as an event is processed.

The **List** shows all events that trigger an action. Click the  button to open it.





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Why Use Simulation?

You can use simulation for planning a new plant or for optimizing an existing plant.

As a rule, you will employ simulation when you have to:

- **Plan a new plant**. Here simulation helps you to:
    - Detect and eliminate problems that otherwise would require cost- and time-consuming correction measures during production ramp-up.
    - Determine and optimize the times, such as processing time, failure time, recovery time, etc., and the throughput of the plant.
    - Determine the size of buffers and the number of machines your intended throughput requires. When a single machine costs hundreds of thousands of dollars, it certainly helps to know if you need one or more machines of one type.
    - Determine the limits of performance of the machines and of the plant as a whole.
    - Investigate how failures affect the throughput and the utilization of the machines.
    - Determine how many workers and staff members are required for the intended throughput.
    - Gain knowledge about the behavior of the plant.
    - Determine suitable control strategies of the machines and of the way the machines interact.
    - Evaluate different alternatives by running a number of simulation experiments.
    - Minimize the investment cost for production lines without jeopardizing required output.
- **Optimize an existing plant**. Here simulation helps you to:
    - Optimize the performance of existing production systems by implementing measures that have been verified in a simulation environment prior to implementation .
    - Optimize the control strategies you devised.
    - Optimize the sequence of orders that have to be fulfilled to make as few tool changes necessary as possible.
    - Test the daily proceedings to make sure that everything works smoothly.
- **Put the plan you formulated into practice**. Here simulation helps you to:
    - Develop a template for creating the control strategies.
    - Test different scenarios during the warm-up phase of the plant.
    - Train the operators of the machines in the different states, which machines and the plant can be in.

In general, you will reap these benefits from employing simulation:

- Enhance the productivity of existing production facilities.
- Reduce investment in planning new production facilities.

- Cut inventory and throughput time.
- Optimize system dimensions, including buffer sizes.
- Reduce investment risks by early proof of concept.
- Maximize use of manufacturing resources.
- Improve line design and schedule.

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Implement a Simulation Project

Developing your simulation model is a cyclical and evolutionary process. You will start out with a first draft of your model and then refine and modify it to make use of the intermediary results the simulation runs provide. Eventually, after several cycles, you will arrive at your final model.

Before you start implementing your simulation project, you will, more or less, proceed like this. You will:

- **Describe the project**

  Determine the goals, so that the purpose of the simulation project becomes clear. Why are you examining a problem? Which questions do you want answered? Put the definition of the project in writing and consult it repeatedly during the course of the project, as the purpose of the simulation study determines the efforts to be made.

- **Plan the project**

  Create a concept of your model, with its initial values, its model items, variables, logic of proceeding and a preliminary description of the simulation experiments. Which parameters do you have to change, which data do you have to collect and how do you interpret this data? Make a list of all functional units that the installation you are modeling will contain. Think about which functional units have identical or similar functionality. Combine them and derive a list of application objects you and your colleagues have to create. Consider re-using existing objects. Specify and plan the remaining objects on paper. Define and describe the interfaces for material and information flow. Outline *reset* and *init methods*.

- **Find out about the data you need and how to acquire it**

  Ensure early on that the data you need to run the simulation experiments is going to be available. Frequently a lot of time and effort is involved to acquire the data. Make sure that you have the name of a person who responsible for acquiring the data from your client, which may, for example, be another department of your company.

- **Build the simulation model**

  Build a first version of the simulation model in its simplest, most basic form. Build the application objects you need and test them one by one. After you are sure that all objects work the way they are supposed to do, put together the overall model. Document the model in a clearly arranged manner, as six months or a year from the time you modeled you might not remember how you accomplished a certain task or why you solved a specific problem the way you did.

- **Verify the simulation model and check its validity**

  After you are finished building the simulation model, you have to verify it, i.e., check if the components you modeled perform the tasks you programmed them to do. Test each and every object you created. Check for the correct functioning and for concurrence with the specifications. Test the objects in combination with other objects and then in the overall model. Make sure that all parameters are set to the correct values. Once you have verified the model, check it for its validity: Make sure the functionality of the model is as expected and conforms to the functionality of the planned or real plant and see if the results are plausible and credible. Make an estimate of the most important results and compare them with the results of the simulation. Introduce your model to a production or planning expert and discuss the results, the proceedings and your modeling approach with him.

- **Execute simulation experiments and collect the results**